Week 5: Advanced Data Manipulation with Pandas

1. Group sales data by month and calculate monthly totals
2. Filter data using multiple conditions (AND, OR operations)
3. Clean and transform text data using string methods
4. Extract year, month, day from date columns
5. Merge customer data with sales data
6. Create pivot tables to summarize data by categories

Project:

Customer Sales Analysis: Analyze customer purchasing patterns, identify top customers, and create a comprehensive sales performance dashboard

Tips & Resources:

- Search YouTube: 'Pandas data manipulation advanced tutorial'
- Practice with real business scenarios
- Learn to think from business perspective
- Build reusable code snippets for common operations
- Document your data transformation steps

In [1]:
```
import pandas as pd
```

1. Group sales data by month and calculate monthly totals

In [2]:
```
df = pd.read_csv("Amazon Store Sales Data.csv")

df['Order Date'] = pd.to_datetime(df['Order Date'])

monthly_sales = df.groupby(df["Order Date"].dt.month)["Sales"].sum().round(1)
monthly_sales.index = df["Order Date"].dt.month_name().str[:3].unique()

monthly_sales = monthly_sales.reset_index()
monthly_sales.columns = ["Month", "Sales"]
print(monthly_sales)
```

```
   Month     Sales
0    Jan   73379.8
1    Feb   72046.9
2    Mar  115945.2
3    Apr   89557.6
4    May  117737.8
5    Jun  109587.3
6    Jul   99339.4
7    Aug  108454.3
8    Sep  193213.7
9    Oct  131584.7
10   Nov  210372.8
11   Dec  244584.9
```

2. Filter data using multiple conditions (AND, OR operations)

In [3]:
```
filtered_and = df[(df["Category"] == "Furniture") & (df["Sales"] > 1000)]
```

```
filtered_or = df[(df["Category"] == "Furniture") | (df["Category"] == "Technology")]
```

3. Clean and transform text data using string methods

In [4]:
```
df["Customer Name"] = df["Customer Name"].str.lower()
df["City"] = df["City"].str.strip()
# df[["First Name", "Last Name"]] = df["Customer Name"].str.split(" ", expand=True)
```

4. Extract year, month, day from date columns

In [5]:
```
df["Year"] = df["Order Date"].dt.year
df["Month"] = df["Order Date"].dt.month
df["Day"] = df["Order Date"].dt.day
```

6. Create pivot tables to summarize data by categories

In [6]:
```
print("----------------------------------------------------------------------")
pivot = pd.pivot_table(
    df,
    values="Sales",
    index="Category",
    columns="Region",
    aggfunc="sum",
    fill_value=0,
)
pivot = pivot.round(2)
print(pivot)

print("----------------------------------------------------------------------")

orders_pivot = pd.pivot_table(
    df,
    index="Month",
    values="Order ID",
    aggfunc="nunique"
).reset_index()

orders_pivot.columns = ["Month", "Order Count"]

print(orders_pivot)
print("----------------------------------------------------------------------")
```

```
----------------------------------------------------------------------
Region              Central       East     South       West
Category
Furniture        105505.45  117071.97  72984.87  155946.35
Office Supplies  150154.35  189066.82  99652.14  204834.37
Technology        85347.72  144095.87  79484.07  161660.33
----------------------------------------------------------------------
    Month  Order Count
0       1          117
1       2           98
2       3          204
3       4          205
4       5          226
5       6          230
6       7          207
7       8          201
8       9          418
9      10          252
10     11          444
11     12          401
----------------------------------------------------------------------
```